

TROY 2 W, AL, USA

WMO: 750010

Lat: 31.7901N

Lon: 86.0004W

Elev: 144

StdP: 99.61

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 23801

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) -2.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 7.4	(k) 18.3	(l) 6.2	(m) 16.1	(n) 1.6	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 34.1	(d) N/A	(e) 33.1	(f) N/A	(g) 32.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.1	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.6	(b) 4.8	(c) 4.2	(e) -8.0	(f) 35.4	(g) 2.8	(h) 1.3	(i) -10.0	(j) 36.3	(k) -11.6	(l) 37.1	(m) -13.1	(n) 37.8	(o) -15.1	(p) 38.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	8.3	11.1	14.2	18.0	22.0	25.6	26.4	26.2	24.3	18.7	12.7	10.7		
	DBStd	7.53	5.96	5.41	4.89	3.61	3.08	1.71	1.55	1.63	2.62	4.32	4.65	5.43		
	HDD10.0	258	104	48	19	1	0	0	0	0	0	2	23	61		
	HDD18.3	1186	312	207	142	50	10	0	0	0	1	49	176	240		
	CDD10.0	3263	53	80	151	241	371	467	509	502	429	271	106	84		
	CDD18.3	1149	2	6	14	40	122	217	251	243	180	61	8	5		
	CDH23.3	9732	2	19	100	344	1066	1982	2212	2045	1444	471	37	8		
	CDH26.7	3849	0	1	9	56	359	880	973	872	583	114	3	0		
(14) Wind	WSAvg	1.8	2.2	2.4	2.4	2.2	1.7	1.6	1.4	1.5	1.4	1.7	1.6	2.1		
(15) Precipitation	PrecAvg	1415	125	123	160	111	92	128	152	114	95	73	106	117		
	PrecMax	2308	272	302	390	233	265	321	396	286	338	310	268	414		
	PrecMin	1031	24	22	10	18	3	26	42	14	16	0	30	31		
	PrecStd	226	59	67	83	61	54	72	78	61	64	69	48	66		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	25.6	28.0	29.8	33.0	35.3	35.0	35.3	34.5	31.4	26.8	24.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	21.5	23.4	26.0	28.4	31.5	33.9	33.6	33.6	33.0	29.4	24.2	22.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	19.8	21.3	24.2	27.0	30.2	32.7	32.6	32.5	31.7	27.8	22.6	20.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	17.7	19.8	22.3	25.5	28.8	31.5	31.6	31.3	30.3	26.1	21.0	19.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	12.3	12.6	12.2	11.1	10.4	10.2	10.9	12.4	12.9	10.5		
		MCDBR	11.3	11.8	13.0	13.6	13.8	13.2	11.7	11.9	12.8	12.9	12.8	10.4		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDWR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.329	0.354	0.381	0.429	0.470	0.520	0.507	0.426	0.368	0.344	0.325		
	taud		2.531	2.511	2.442	2.366	2.267	2.209	2.077	2.113	2.320	2.458	2.470	2.533		
	Ebn at Noon		906	932	927	910	863	824	781	787	843	875	868	882		
	Edh at Noon		85	95	109	123	137	144	164	157	122	99	90	81		
(44) All-Sky Solar Radiation	RadAvg		2.90	3.47	4.65	5.75	6.31	6.12	5.97	5.53	4.95	4.26	3.27	2.50		
	RadStd		0.19	0.43	0.40	0.46	0.44	0.48	0.38	0.34	0.39	0.51	0.29	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+165	+107

Nomenclature: See separate page